

Week 13 — Linear Time-Invariant (LTI) Filters

21 May 2026

The big picture. An LTI filter acts as a convolution (impulse response) and multiplies the spectrum by $|H(f)|^2$ (transfer function); hence the ARMA spectral density.

A *filter* turns one sequence into another, $\{Y_t\} = \mathcal{L}[\{X_t\}]$. The filters that matter for time series are the **linear time-invariant** (LTI) ones: they treat every time point the same and respect superposition. Two “test inputs” pin such a filter down completely — a single impulse (giving the *impulse response* h) and a pure wave (giving the *transfer function* H). The payoff is one formula, $\mathcal{S}_Y(f) = |H(f)|^2 \mathcal{S}_X(f)$, which is the engine behind the closed-form ARMA spectral density. (We allow complex-valued series and assume the sums converge, the usual sufficient condition being $h \in \ell^1$.)

Key definitions

Definition — LTI filter

A digital filter $\mathcal{L}$ (with output entry $Y_u = \mathcal{L}[\{X_t\}]_u$) is **linear time-invariant** if for all sequences and all $\alpha \in \mathbb{C}$:

1. **Linearity:** $\mathcal{L}[\{\alpha X_t + Y_t\}] = \alpha \mathcal{L}[\{X_t\}] + \mathcal{L}[\{Y_t\}]$;
2. **Time invariance:** $\mathcal{L}[B[\{X_t\}]] = B[\mathcal{L}[\{X_t\}]]$ — $\mathcal{L}$ commutes with the backshift B .

The one-step condition already gives $\mathcal{L}[B^u\{X_t\}] = B^u \mathcal{L}[\{X_t\}]$ for all u . The backshift B ($B[\{X_t\}]_u = X_{u-1}$) is itself the prototypical LTI filter.

Definition — Impulse response h

The **impulse sequence** is $\delta_{t,m} = 1$ if $t = m$, else 0. The **impulse response** of $\mathcal{L}$ is

$$h_m = \mathcal{L}[\{\delta_{t,-m}\}]_0, \quad m \in \mathbb{Z}$$

(the 0 is the output time). Informally: the response, read at time 0, to a unit shock placed at time $-m$. Regularity condition: $h \in \ell^1$.

Definition — Transfer function H

For $f \in \mathbb{R}$ the **complex wave** is $\xi_{t,f} = e^{2\pi i f t}$. The **transfer function** of $\mathcal{L}$ is

$$H(f) = \mathcal{L}[\{\xi_{t,f}\}]_0, \quad f \in \mathbb{R},$$

the filter’s frequency-domain fingerprint: how each pure frequency is re-weighted in amplitude and shifted in phase.

Two test signals, two views of the same filter

Feed in a *spike* $\Rightarrow$ you read off the time-domain weights h (the convolution kernel). Feed in a *pure wave* $\Rightarrow$ you read off the frequency-domain gain $H(f)$. Same filter, two complementary descriptions, linked by a Fourier transform.

Key results & formulas

Theorem — LTI $\Leftrightarrow$ convolution with h

Under convergence assumptions, an LTI filter acts as a **convolution** with its impulse response:

$$\mathcal{L}[\{X_t\}]_u \stackrel{\text{ms}}{=} \sum_{m \in \mathbb{Z}} h_{u-m} X_m, \quad u \in \mathbb{Z},$$

and conversely any convolution with $h \in \ell^1$ is LTI. *Idea:* write the input as a sum of impulses, use linearity on the truncations and time invariance ($\mathcal{L}[\{\delta_{t,m}\}]_u = h_{u-m}$); absolute convergence passes to the full series. This is the master representation: any LTI filter = the weights h .

Theorem — waves are eigensequences, $H(f)$ the eigenvalue

For every f , $\mathcal{L}[\{\xi_{t,f}\}] = H(f) \{\xi_{t,f}\}$: the filter only rescales amplitude and shifts phase, never changing the frequency. Hence in the frequency basis filtering is **pointwise multiplication** by $H(f)$. *Idea:* $B^{-u} \{\xi_{t,f}\} = e^{2\pi i u f} \{\xi_{t,f}\}$, then push B^u through $\mathcal{L}$ using time invariance.

Theorem — H is the Fourier transform of h

If $h \in \ell^1$,

$$H(f) = \sum_{m \in \mathbb{Z}} h_m e^{-2\pi i f m}, \quad f \in \mathbb{R}.$$

Recover the weights as Fourier coefficients: $h_m = \int_{-1/2}^{1/2} H(f) e^{2\pi i f m} df$. *Practical recipe:* read h off the filter, sum the series to get H .

Theorem — filtering preserves stationarity

If $\{X_t\}$ is stationary and $h \in \ell^1$, then $\{Y_t\} = \mathcal{L}[\{X_t\}]$ is stationary: $\mathbb{E}[Y_t] = \mu_X \sum_m h_m$ is t -free, and $\gamma_\tau^{(Y)} = \sum_{m,s} h_m \bar{h}_s \gamma_{\tau-m+s}^{(X)}$ depends only on τ (with $\gamma_0^{(Y)} \leq \gamma_0^{(X)} \|h\|_1^2 < \infty$). Only then does S_Y even exist.

Theorem — the workhorse: $S_Y(f) = |H(f)|^2 S_X(f)$

For $\{X_t\}$ stationary with $\gamma^{(X)} \in \ell^1$ and $\mathcal{L}$ LTI with $h \in \ell^1$,

$$S_Y(f) = |H(f)|^2 S_X(f), \quad f \in \mathbb{R}.$$

Idea: $\gamma^{(Y)} = w * \gamma^{(X)}$ with $w = h * \tilde{h}$, $\tilde{h}_t = \overline{h_{-t}}$; the convolution theorem gives the multiplier $W(f) = H(f) \overline{H(f)} = |H(f)|^2$. The gain is the **squared modulus** — always real and non-negative, as a spectral density must be.

Lemma + Theorem — polynomial filter $\Rightarrow$ ARMA spectrum

A polynomial filter $\Phi(B) = \sum_{j=0}^p \phi_j B^j$ is LTI with $H(f) = \Phi(e^{-2\pi i f})$. Applying the workhorse to both sides of $\Phi(B)X_t = \Theta(B)\varepsilon_t$ (white noise, variance σ^2) gives the **ARMA(p, q) spectral density**

$$S_X(f) = \sigma^2 \frac{|\Theta(e^{-2\pi i f})|^2}{|\Phi(e^{-2\pi i f})|^2}.$$

Division is legal because stationarity forbids zeros of Φ on the unit circle. ARMA has *no* closed-form ACVS, but this clean closed-form spectrum.

Filtering is “multiply the spectrum by a gain”

In the time domain filtering is a (sometimes messy) convolution. In the frequency domain it collapses to multiplication by $|H(f)|^2$. So a filter is just a frequency-by-frequency volume knob: $|H(f)|^2 > 1$ amplifies that band, < 1 attenuates it, $= 0$ kills it. Where H is large the output spectrum bulges; where $H = 0$ the output is silent.

Why ARMA is “MA on top, AR on the bottom”

A cascade $\mathcal{L}_1\mathcal{L}_2$ multiplies transfer functions, $H = H_1H_2$ (convolution in time = product in frequency). ARMA is the MA filter $\Theta(B)$ acting on white noise, then “undone” by the AR filter $\Phi(B)$: the MA part contributes $|\Theta|^2$ as a numerator, the AR part $|\Phi|^2$ as a denominator. Roots of Θ near the circle dig dips; roots of Φ near the circle raise peaks.

First difference $Y_t = X_t - X_{t-1}$ is high-pass

The filter $I - B$ has impulse response $h_0 = 1, h_1 = -1$, so

$$H(f) = 1 - e^{-2\pi if}, \quad |H(f)|^2 = |1 - e^{-2\pi if}|^2 = 4 \sin^2(\pi f).$$

Therefore $\mathcal{S}_Y(f) = 4 \sin^2(\pi f) \mathcal{S}_X(f)$: differencing *kills* $f = 0$ (since $H(0) = 0$, it removes trend/DC) and *amplifies* high frequencies — a high-pass operation. (Handy identity: $|1 - e^{-2\pi if}|^2 = 2 - 2 \cos(2\pi f) = 4 \sin^2(\pi f)$.)

Key takeaways

Key takeaways — the week’s reflexes

To get the spectrum of a filtered process $\{Y_t\} = \mathcal{L}[\{X_t\}]$:

1. **Find** h (the convolution weights), or write $\mathcal{L}$ as a polynomial $\Phi(B)$. To read h off a convolution: in $\sum_k h_k X_{u-k}$ the coefficient of X_{u-k} is h_k .
2. **Form** $H(f) = \sum_m h_m e^{-2\pi ifm}$ (for $\Phi(B)$ this is just $\Phi(e^{-2\pi if})$).
3. **Multiply:** $\mathcal{S}_Y(f) = |H(f)|^2 \mathcal{S}_X(f)$.

Reflexes:

- Prove LTI by checking the two axioms; or note that any linear combination / cascade of LTI filters, and any convolution with $h \in \ell^1$, is automatically LTI.
- **Cascades multiply:** $H = H_1H_2$, gains $|H_1|^2 |H_2|^2$.
- The multiplier is $|H(f)|^2$, *not* $H(f)$ or $H(f)^2$.
- **Complex ACVS:** the conjugate sits on the *second* factor, $\text{Cov}(X_{t+\tau}, X_t) = \mathbb{E}[(X_{t+\tau} - \mu)(\overline{X_t - \mu})]$.
- Filtering preserves stationarity ($h \in \ell^1$) — a prerequisite for $\mathcal{S}_Y$ to exist.
- AR(1) peaks at $f = 0$ if $\phi > 0$ (persistent), at $f = \frac{1}{2}$ if $\phi < 0$ (oscillating).